

DocLens: Software Requirements Specification (SRS)

ISO/IEC/IEEE 29148 & IEEE 830 Standard Product Engineering Specification

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1. Introduction & Document Purpose

1.1 Purpose of Document

This Software Requirements Specification (SRS) defines the functional, non-functional, system interface, data architecture, and security requirements for DocLens v1.0.0. This specification serves as the formal technical agreement for system engineers, QA testers, security auditors, and hackathon judges.

1.2 Scope of System

DocLens is a multi-tier web application that automates signature verification from raw document uploads (images and multi-page PDFs). The platform performs automatic signature region localization, 512-dimensional deep neural feature extraction via PyTorch ResNet18, 5-point classical structural stroke measurements via OpenCV, calibrated score calculation based on frozen writer-disjoint CEDAR benchmarks, Grad-CAM visual difference heatmap generation, LLM narrative report synthesis, and vector PDF export.

1.3 Definitions & Acronyms

- **CEDAR:** Center of Excellence for Document Analysis and Recognition dataset.
- **Grad-CAM:** Gradient-weighted Class Activation Mapping explainability visual layer.
- **RLS:** Row-Level Security policy enforced at the PostgreSQL database layer.
- **FAR / FRR / EER:** False Acceptance Rate, False Rejection Rate, and Equal Error Rate.

2. Overall System Description

2.1 Product Architecture

DocLens comprises three primary technical tiers: a React 18 single-page application (SPA) frontend built with Vite 6; an asynchronous FastAPI Python 3.11 backend serving OpenCV, PyTorch, PyMuPDF, and Grad-CAM modules; and Supabase cloud infrastructure managing JWT authentication, PostgreSQL relational data, and isolated file storage.

2.2 User Classes & Roles

- **Forensic Document Examiners:** Require granular structural metrics, baseline trajectory data, pen-lift coordinates, and raw embedding distance scores.
- **Compliance & Banking Auditors:** Require clean executive verdict summaries, calibrated confidence indicators, and exportable PDF audit artifacts.
- **Technical Judges & Developers:** Require OpenAPI compliance, RESTful API documentation, reproducible benchmark scripts, and robust error handling.

3. Specific System Requirements (Exhaustive Functional Requirements)

3.1 User Authentication & Session Management (FR-1)

- **FR-1.1 Account Registration:** The system shall allow new users to register an account using a valid email address and password.
- **FR-1.2 JWT Token Authentication:** The system shall issue signed JWT bearer tokens upon successful user login.
- **FR-1.3 Route Protection:** The system shall reject unauthenticated HTTP requests to API endpoints under /api/v1/cases and /api/v1/analyze with HTTP 401 Unauthorized.

- **FR-1.4 Password Recovery:** The system shall support password recovery via email reset tokens handled securely through Supabase Auth.
- **FR-1.5 Authenticated Profile:** The system shall provide a GET /api/v1/auth/me endpoint returning current authenticated user details.

3.2 Case Workspace Management (FR-2)

- **FR-2.1 Case Creation:** Users shall be able to create new signature verification cases specifying Case Title and Signer Name.
- **FR-2.2 Multi-Case Dashboard:** The system shall display a dashboard listing all cases owned by the authenticated user.
- **FR-2.3 Case Metadata Tracking:** Each case shall track Case ID, Title, Signer Name, Status (draft, analyzed, archived), Creation Date, and Document Count.
- **FR-2.4 Case Deletion:** Deleting a case shall cascade-delete all associated document records and stored storage objects in cloud storage.
- **FR-2.5 Case Filtering:** The dashboard UI shall support search and filtering of cases by title, signer name, or analysis status.

3.3 Document & Specimen Ingestion Engine (FR-3)

- **FR-3.1 Format Support:** The system shall ingest document uploads in JPEG, JPG, PNG, WebP, BMP, and PDF file formats.
- **FR-3.2 Multi-Reference Support:** The system shall accept 1 questioned document and 1 to 5 genuine reference specimen signatures per case.
- **FR-3.3 File Validation:** System shall validate MIME types and reject corrupted or unparseable upload binary payloads before passing them to the pipeline.
- **FR-3.4 Size Validation:** System shall enforce a strict 10 MB maximum file size cap per uploaded document.

3.4 Automated Signature Localization (FR-4)

- **FR-4.1 Bounding Box Detection:** The OpenCV detection engine shall automatically locate signature region candidates without manual user cropping.
- **FR-4.2 Candidate Region Scoring:** Candidate bounding boxes shall be scored based on aspect ratio, area density, ink compactness, connected component structure, and text penalties.
- **FR-4.3 Bounding Box Persistence:** Detected signature coordinates shall be retained in original document coordinate space for evidence visualization.

3.5 PDF Processing & Candidate Selection (FR-5)

- **FR-5.1 Page Rendering:** PyMuPDF shall render PDF pages at 200 DPI within pixel safety limits.
- **FR-5.2 Candidate Inspection:** The system shall inspect text density and extract embedded images to select the optimal page representation for signature analysis.
- **FR-5.3 Encrypted PDF Rejection:** System shall reject password-protected or encrypted PDF documents returning HTTP 422 Unprocessable Entity.
- **FR-5.4 Page Cap Enforcement:** System shall limit multi-page PDF processing to a maximum of 40 pages per document upload.

3.6 Deep Visual Feature Extraction (FR-6)

- **FR-6.1 ResNet18 Feature Trunk:** The system shall pass normalized 224x224 RGB signature crops through a pretrained ResNet18 feature extractor operating in PyTorch

inference_mode().

- **FR-6.2 512D L2 Embeddings:** The model shall output a 512-dimensional L2-normalized vector embedding for each signature.

3.7 Multi-Reference Comparison Engine (FR-7)

- **FR-7.1 Cosine Similarity Calculation:** The system shall compute normalized cosine similarity between questioned embeddings and all reference specimen embeddings.
- **FR-7.2 Reference Profile Centroid:** The system shall measure similarity against the centroid vector of the genuine reference set.
- **FR-7.3 Consensus Spread Measurement:** System shall evaluate similarity score spread across all reference specimens to measure signer variability.

3.8 Classical Structural Feature Extraction (FR-8)

- **FR-8.1 Stroke Width Variation:** System shall measure stroke thickness distribution using distance transforms.
- **FR-8.2 Letter Proportion Inconsistency:** System shall evaluate connected component aspect ratios.
- **FR-8.3 Baseline Deviation:** System shall calculate writing baseline orientation trajectory.
- **FR-8.4 Pen Lift Coordinates:** System shall map spatial locations of likely stroke gaps.
- **FR-8.5 Slant Angle Deviation:** System shall estimate principal rotational axes.

3.9 Calibrated Assessment Engine (FR-9)

- **FR-9.1 Ensemble Score Calculation:** Final score shall combine 95% Deep Embedding Similarity and 5% Classical Structural Similarity.
- **FR-9.2 CEDAR Threshold Evaluation:** Scores ≥ 0.9271469 shall be classified as "AUTHENTIC"; scores < 0.9271469 as "QUESTIONED / DISCREPANCY".

3.10 Grad-CAM Visual Evidence Overlay (FR-10)

- **FR-10.1 Heatmap Generation:** System shall generate pixel-level Grad-CAM feature activation maps from the final convolutional layer of ResNet18.
- **FR-10.2 HTML5 Canvas Rendering:** The frontend shall overlay Grad-CAM heatmaps, bounding boxes, baseline indicators, and evidence markers on document canvas views.

3.11 Generative AI Narrative Engine (FR-11)

- **FR-11.1 Groq Llama-3.3 Integration:** Structured numerical findings shall be sent to Groq API (llama-3.3-70b-versatile).
- **FR-11.2 Guardrail Enforcement:** Prompt guardrails shall strictly prohibit changing the calculated verdict, inventing non-existent evidence, or claiming legal absolute certainty.

3.12 Deterministic Fallback Engine (FR-12)

- **FR-12.1 Fallback Activation:** If Groq API times out or fails, the system shall generate a structured template report preserving all calculated scores.

3.13 Interactive Web Dashboard (FR-13)

- **FR-13.1 React UI Layout:** React client shall present Case Summary, Interactive Canvas, Structural Metric Cards, and Report Summary tabs.

3.14 PDF Report Exporter (FR-14)

- **FR-14.1 jsPDF Export Engine:** Frontend shall generate standalone vector PDF case reports containing evidence images, feature tables, narrative text, and verification QR codes.

4. Non-Functional Requirements (NFR)

Requirement Area	Quantitative Target / Safety Threshold	Enforcement Method
NFR-1 Execution Latency	Complete end-to-end analysis in < 3.5 seconds	PyTorch inference_mode() & Uvicorn async
NFR-2 Upload Size Cap	10 MB maximum per uploaded file	FastAPI HTTP 413 file length check
NFR-3 Pixel Safety Limit	20 Million maximum decoded pixels	Pillow image size validation check
NFR-4 PDF Page Limit	40 Maximum pages per PDF upload	PyMuPDF page count validation
NFR-5 Security & RLS	Strict multi-tenant PostgreSQL data isolation	Supabase Auth JWT claims (auth.uid())
NFR-6 Benchmark Acc	>= 95.0% Balanced Accuracy on test split	CEDAR writer-disjoint validation split

5. Database Schema & Data Dictionary

The relational database schema is implemented in Supabase PostgreSQL with full foreign-key cascading:

- **Table: cases** id UUID PRIMARY KEY DEFAULT gen_random_uuid(), user_id UUID NOT NULL REFERENCES auth.users(id) ON DELETE CASCADE, title TEXT NOT NULL, signer_name TEXT NOT NULL, status TEXT NOT NULL DEFAULT 'draft', created_at TIMESTAMPTZ DEFAULT now(), updated_at TIMESTAMPTZ DEFAULT now().
- **Table: documents** id UUID PRIMARY KEY DEFAULT gen_random_uuid(), case_id UUID NOT NULL REFERENCES public.cases(id) ON DELETE CASCADE, file_type TEXT NOT NULL, file_name TEXT NOT NULL, storage_path TEXT NOT NULL, created_at TIMESTAMPTZ DEFAULT now().
- **Table: analysis_results** id UUID PRIMARY KEY DEFAULT gen_random_uuid(), case_id UUID NOT NULL REFERENCES public.cases(id) ON DELETE CASCADE, verdict TEXT NOT NULL, confidence_score FLOAT NOT NULL, deep_similarity FLOAT NOT NULL, structural_score FLOAT NOT NULL, features_json JSONB NOT NULL, narrative_report TEXT NOT NULL, created_at TIMESTAMPTZ DEFAULT now().

6. Security Architecture & RLS Isolation

Multi-tenant data security is enforced at the database and storage layers using PostgreSQL Row-Level Security (RLS) policies. Every table query evaluates auth.uid() against the user_id column:

CREATE POLICY "Users can access own cases" ON public.cases FOR ALL USING (auth.uid() = user_id);

CREATE POLICY "Users can access own documents" ON public.documents FOR ALL USING (case_id IN (SELECT id FROM public.cases WHERE user_id = auth.uid()));

CREATE POLICY "Users can access own analysis" ON public.analysis_results FOR ALL USING (case_id IN (SELECT id FROM public.cases WHERE user_id = auth.uid()));

7. Verification & Acceptance Testing Matrix

Requirement ID	Verification Method	Acceptance Criteria
FR-1 & NFR-5	Security Audit	Unauthenticated HTTP requests return 401; users cannot access other users' cases.
FR-4 & FR-5	Integration Test	Multi-page PDF upload correctly isolates signature bounding box without manual crop.
FR-6 & FR-9	Benchmark Test	Model achieves 96.50% balanced accuracy on unseen test writers using 0.9271469 threshold.
FR-10	Visual Inspection	Grad-CAM heatmaps render correctly over detected signature coordinates on HTML5 canvas.
FR-11 & FR-12	Fault Tolerance Test	System outputs Groq LLM summary when API is active; switches cleanly to template engine when offline.
FR-14	PDF Validation	Exported PDF opens cleanly in Adobe Acrobat and contains valid verification QR code.

8. System Reliability & Maintenance Standards

DocLens is configured for continuous zero-downtime deployment on Render cloud infrastructure. The backend FastAPI service includes automated health check monitoring via `/api/v1/health`, graceful shutdown handlers, and worker thread pool isolation to ensure PyTorch inference tasks never block HTTP event loops.

In addition, file uploads are automatically validated against MIME signatures before being processed, preventing malicious payload injection or arbitrary binary code execution vectors.

9. Compliance & Data Erasure Protocols

To comply with enterprise privacy requirements and GDPR Right to Erasure mandates, deleting a case triggers automated cascade deletion across PostgreSQL document metadata records and removes corresponding raw storage objects from Supabase Cloud Storage.